

Solutions to JEE Advanced - 8 | JEE 2022 | Paper - 2

PHYSICS

$$1.(AC) \quad f = \left(\frac{V_s + V}{V_s} \right) f_0 \Rightarrow f = \left(\frac{V_s + gt}{V_s} \right) f_0$$

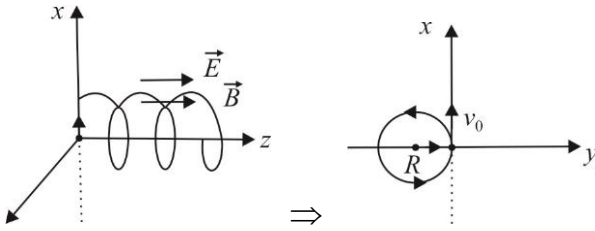
$$\Rightarrow \quad f = f_0 + \frac{f_0 gt}{V_s}$$

Hence it is straight line

$$2000 = 1000 \left(\frac{V_s + 10 \times 30}{V_s} \right)$$

$$\Rightarrow \quad V_s = 300 \text{ m/s}$$

2.(AB)



$$qV_0 B = \frac{mV_0^2}{R}$$

$$R = \frac{mV_0}{qB} \quad \text{at } t = \frac{\pi m}{qB}, y = \frac{-2mV_0}{qB}$$

$$z = ut + \frac{1}{2}at^2 = 0 + \frac{qE}{m}t^2 \quad \left[T = \frac{2\pi m}{qB} \right]$$

$$z_1 = \frac{qE}{m}T^2$$

$$z_2 = \frac{qE}{m}(2T)^2$$

$$3.(AC) \quad \frac{mv^2}{r} = qvB \Rightarrow v = \frac{qBr}{m} = 5.27 \times 10^5 \text{ m/s}; \quad m = 9.1 \times 10^{-31} \text{ kg}$$

$$\text{Kinetic energy of electron} = \frac{1}{2}mv^2 = 126.3 \times 10^{-21} \text{ J} = 0.8 \text{ eV}; \quad q = 1.9 \times 10^{-19} \text{ C}$$

The energy of the emitted photon when electron jumps from $n = 3$ to $n = 2$; $r = 10 \times 10^{-3} \text{ m}$

$$= \left(-\frac{13.6}{3^2} \right) - \left(-\frac{13.6}{2^2} \right) = 1.89 \text{ eV}$$

$$\text{Work function} = E - K_{\max} = 1.89 - 0.8 = 1.09 \text{ eV}$$

$$4.(AC) \quad \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{u-v}{uv} = \frac{1}{f}$$

$$uv = 100f = 1600$$

$$\Rightarrow u = 20cm \text{ and } 80cm \text{ (} v = 80cm \text{ and } 20cm \text{ respectively)}$$

5.(AD) Dipole moment of the two bigger arcs is

$$\int dP_1 = \int_{-\pi/4}^{\pi/4} dq (2\sqrt{3}R) \cos \theta$$

$$\int dP_1 = \int_{-\pi/4}^{\pi/4} 6\lambda R^2 \cos \theta d\theta \quad (\because dq = (\sqrt{3}R)\lambda d\theta)$$

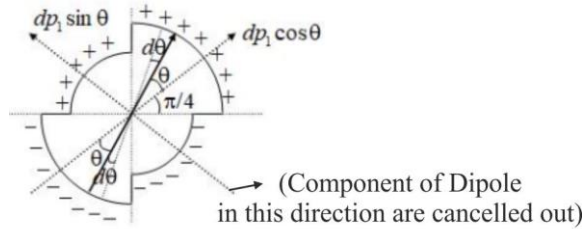
$$P_1 = 6\lambda R^2 \int_{-\pi/4}^{\pi/4} \cos \theta d\theta$$

$$= 6\sqrt{2}\lambda R^2$$

Similarly for smaller arcs the dipole moment is

$$P_2 = 2\sqrt{2}\lambda R^2$$

$$|\vec{P}_{net}| = \sqrt{P_1^2 + P_2^2} = 4\sqrt{5}\lambda R^2$$



$$6.(ABC) \quad D = 1m, \quad d = 5 \times 10^{-3}m, \quad \lambda = 500 \times 10^{-9}m, \quad t_1 = 2 \times 10^{-3}m, \quad t_2 = 1.5 \times 10^{-3}m$$

$$\text{Shift due to the first slab, } y_1 = (\mu_1 - 1) \frac{t_1 D}{d} = 20cm$$

$$\text{Shift due to the } 2^{nd} \text{ slab, } y_2 = (\mu_2 - 1) \frac{t_2 D}{d} = 12cm$$

$$\text{Net shift} = 20 - 12 = 8cm$$

Shift will be towards the 1st slab.

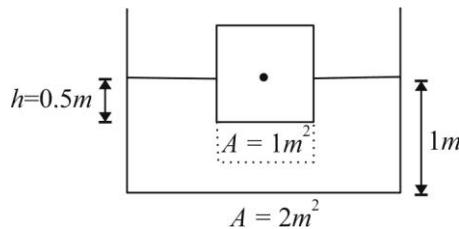
$$7.(C) \quad mg = \rho_l V_{sub} g$$

$$\rho_0 V_0 = \rho_l (1 \times h)$$

$$\rho_0 (1) = \rho_l \times h$$

As cube is pushed, water rises

$$\frac{500}{1000} = \frac{h}{1} \Rightarrow h = 0.5m$$



If it is pushed by x , water level rises by x . So it is submerged by $2x$.

To dip completely, we need to push it by $0.25m$ only. Then buoyant force becomes constant.

$$8.(C) \quad W_{man} + W_g + W_B = 0 \quad (\text{by WET})$$

$$W_{man} + 500 \times 10 \times 0.5 + W_B = 0$$

$$W_B = W_{gravity \text{ on liq.}}$$

$$W_B = - \left[\int_0^{0.25} 10^3 \times g \times (0.5 + 2x) dx + \int_{0.25}^{0.5} 10^3 \times g \times (dx) \right]$$

$$W_B = -10^4 \left[0.5 \times 0.25 + 0.25^2 + (0.5 - 0.25) \right]$$

$$W_B = -10^4 \left[\frac{1}{8} + \frac{1}{16} + \frac{1}{4} \right] = \frac{-(2+1+4) \times 10^4}{16} = \frac{-7}{16} \times 10^4$$

$$\therefore W_{man} = 10^4 \left[\frac{7}{16} - \frac{1}{4} \right] = \frac{3}{16} \times 10^4 = 1875 J$$

9.(C) $\int \vec{E} \cdot d\vec{l} = \frac{d\phi}{dt} \Rightarrow E \cdot 2\pi R = \pi R^2 \cdot 2B_0 t$ (E: induced electric field)

$$\Rightarrow E = B_0 R t; \quad \tau_B = \int (dQE) R = QB_0 R^2 t$$

Torque due to magnetic field should be just equal to torque due to friction.

$$\tau_f = \mu mgR = \left(\frac{QB_0 R}{mg} \right) mgR = QB_0 R^2$$

$$\tau_B = \tau_f \Rightarrow QB_0 R^2 t = QB_0 R^2 \Rightarrow t = 1s$$

10.(D) Angular acceleration after $t = 1s$

$$I\alpha = \tau_B - \tau_f$$

$$(mR^2)\alpha = QB_0 R^2 t - QB_0 R^2$$

$$\alpha = \frac{QB_0}{m} (t-1)$$

$$\int_0^w d\omega = \frac{QB_0}{m} \int_1^t (t-1) dt$$

$$\omega = \frac{QB_0}{m} \left[\frac{t^2}{2} - t + \frac{1}{2} \right]$$

$$\text{At } t = 3s, \omega = \frac{2QB_0}{m}$$

11.(D), 12.(C)

$$w = w_{12} + w_{23} + w_{31} \quad [2 \rightarrow 3, (3P_0)(3V_0) = P_0 \times V]$$

$$= \frac{1}{2} (4P_0)(2V_0) + nRT \ln(3) - P_0(8V_0)$$

$$= -4P_0V_0 + 9P_0V_0 \ln 3$$

$$= (9 \ln 3 - 4) P_0V_0$$

$$C = \frac{\Delta Q}{n\Delta T} = \frac{n \left(\frac{3R}{2} \Delta T \right) + \Delta w}{n\Delta T}$$

$$= \frac{3R}{2} + \frac{4P_0V_0}{n\Delta T} \quad (\because 8P_0V_0 = nR\Delta T \text{ from } 1 \rightarrow 2)$$

$$= \frac{3R}{2} + \frac{P_0V_0}{n(2T_0)}$$

$$= \frac{3R}{2} + \frac{R}{2} = 2R$$

$$\Delta Q_{123} = n(2R)(\Delta T_{12}) + (w_{23})$$

$$= n(2R)(8T_0) + nR(9T_0)\ln(3)$$

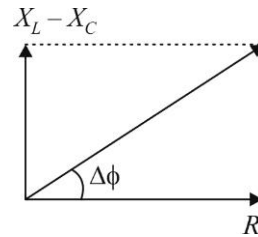
$$= nRT_0(16 + 9\ln 3)$$

$$\eta = \frac{w}{\Delta Q_{123}} = \left(\frac{9\ln 3 - 4}{9\ln 3 + 16} \right)$$

$$1.(0) \quad \frac{\pi}{4} = \tan^{-1} \left(\frac{X_L - X_C}{R} \right) \Rightarrow R = X_C$$

$$X_C' = 4X_C \quad X_L' = 2X_L = 4X_C \quad R' = 2R$$

$$\phi' = \tan^{-1} \left(\frac{X_L' - X_C'}{R'} \right) = 0$$



2.(126) For just submerged, densities should be equal

$$\text{or} \quad (200 + 10)g = \rho_w V g \text{ at } 20^\circ C$$

at new temperature,

$$(200 + 10 - \Delta m)g = \left(\frac{\rho_w}{1 + \gamma \Delta T} \right) [V(1 + 3\alpha \Delta T)]g$$

$$\rho_w V - \Delta m = \rho_w V \frac{(1 + 3\alpha \Delta T)}{(1 + \gamma \Delta T)}$$

Binomial approximation,

$$= \rho_w V [1 + (3\alpha - \gamma) \Delta T]; \quad \Delta m = \rho_w V (\gamma - 3\alpha) \Delta T$$

$$\Delta m = (210) [1.5 \times 10^{-4} - 3 \times 10^{-5}] \times 5 \Rightarrow \Delta m = 126 \text{ gm}$$

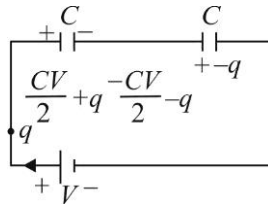
3.(8) Let charge supplied by battery is q .

Apply KVL

$$\frac{CV}{2} + q + \frac{q}{C} = V$$

$$\frac{CV}{2} + 2q = CV$$

$$2q = \frac{CV}{2}; q = \frac{CV}{4}$$



$$\text{Energy supplied by battery} = q \times V = \frac{CV^2}{4}; \text{Energy consumed by capacitors} = U_f - U_i$$

$$= \left[\frac{1}{2} C \left(\frac{3V}{4} \right)^2 + \frac{1}{2} C \left(\frac{V}{4} \right)^2 \right] - \left[\frac{1}{2} C \left(\frac{V}{2} \right)^2 + 0 \right] = \frac{9CV^2}{32} + \frac{CV^2}{32} - \frac{CV^2}{8} = \frac{3CV^2}{16}$$

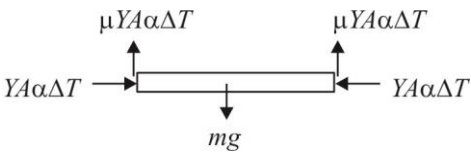
$$\text{Heat loss} = \frac{CV^2}{4} - \frac{3CV^2}{16} = \frac{CV^2}{16} = \frac{CV^2}{2 \times 8}$$

4.(6) $N_1 = 2N_2$

$$N_0 e^{-\lambda_1 t} = 2N_0 e^{-\lambda_2 t}$$

$$\Rightarrow \frac{1}{t} = \frac{1}{T_2} - \frac{1}{T_1}$$

5.(25)



$$mg = 2\mu YA\alpha\Delta T \Rightarrow \Delta T = \frac{\rho Lg}{2\mu Y\alpha}$$

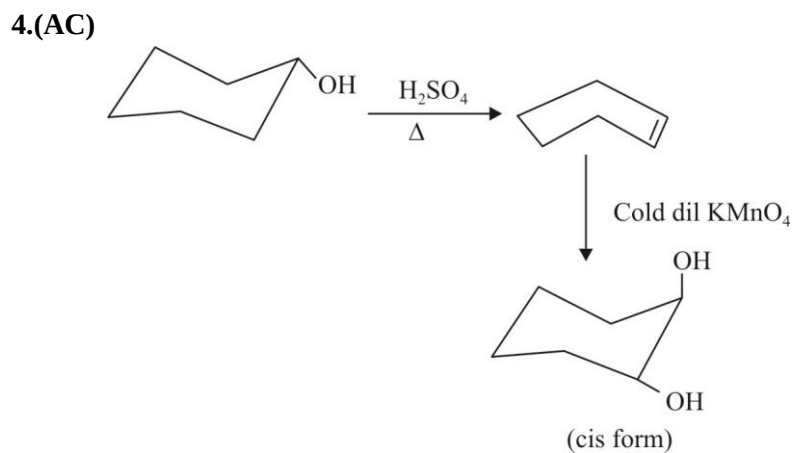
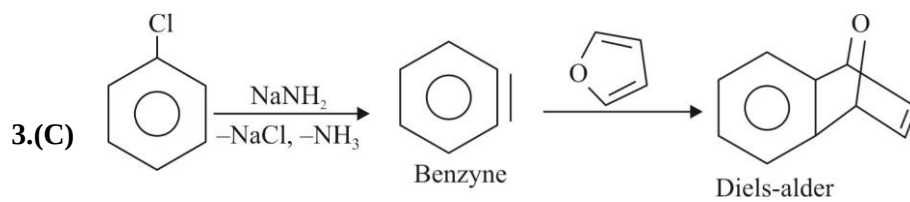
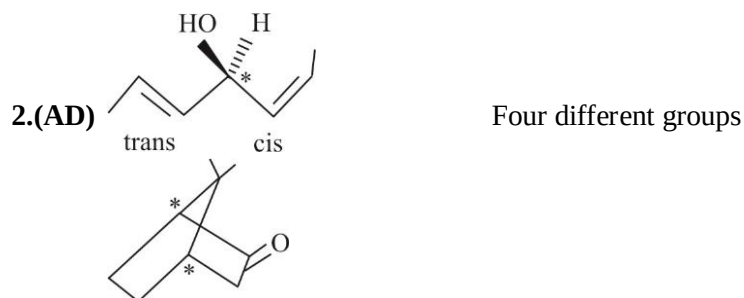
$$\Rightarrow \Delta T = \frac{50 \times 10^3 \times 8 \times 10}{2 \times 0.2 \times 2 \times 10^{11} \times 2 \times 10^{-6}} = 25^\circ\text{C}$$

6.(4) $t = \frac{T}{4} = \frac{2\pi}{4} \sqrt{\frac{I}{PE}}$

$$I = \frac{4t^2}{\pi^2} pE$$

CHEMISTRY

- 1.(CD) (A) False $E^0_{F_2/F^-} = \text{maximum}$ so can not be oxidized.
 (B) False: $\text{HCl}_{(\text{aq})} \longrightarrow \text{H}^+ + \text{Cl}^-$ aq HCl is good conductor due to ions.
 (C) True : $\text{H}_2\text{O} + \text{Cl}_2 \longrightarrow 2\text{HCl} + (\text{O})$ nascent oxygen
 (D) True: Chlorine being more reactive displaces iodine from KI. Iodine is absorbed by starch and thus blue colour is developed
 $2\text{KI} + \text{Cl}_2 \longrightarrow 2\text{KCl} + \text{I}_2$
 $\text{I}_2 + \text{Starch} \longrightarrow \text{Blue colour}$

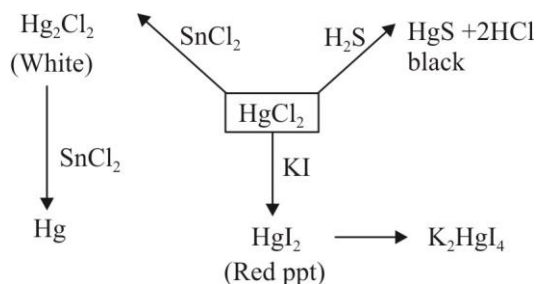


5.(CD) Highest reducing power of Li is due to large heat of hydration.

6.(CD)

7.(C)

8.(C)



9.(C) EMF to be positive, the correct representation of the cell (a concentration cell) will be as follows:

$\text{Pt}, \text{H}_2 (15 \text{ atm}) | \text{HCl} | \text{H}_2 (3 \text{ atm}), \text{Pt}$ Oxidation at LHE:

$\text{H}_2 (15 \text{ atm}) \longrightarrow 2\text{H}^+ + 2\text{e}^-$ (Oxidation)

Reduction at RHE :

$2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2 (3 \text{ atm})$ (Reduction)

Overall change:

$\text{H}_2 (15 \text{ atm}) \longrightarrow \text{H}_2 (3 \text{ atm})$

EMF of a concentration cell is given by the expression

$$E_{\text{cell}} = \frac{0.059}{n} \log \frac{C_2}{C_1} = \frac{0.059}{2} \log \frac{P_2}{P_1}, P_2 > P_1$$

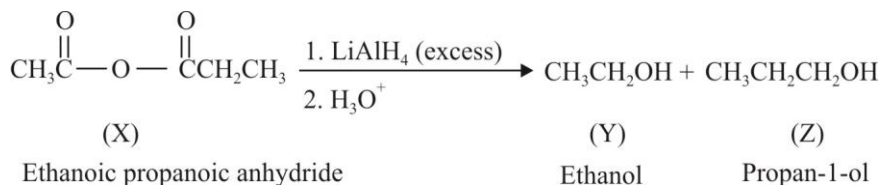
$$= \frac{0.059}{2} \log \frac{15}{3} = +0.0206\text{V}$$

However, for the cell given: $E_{\text{cell}} = -0.0206\text{V}$

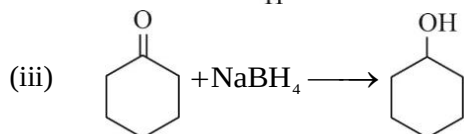
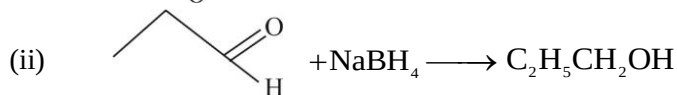
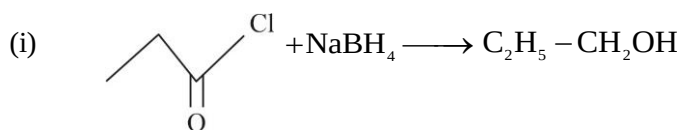
10.(B) $E_{\text{cell}} = E_{\text{cell}}^0 - \frac{RT}{nF} \ln \frac{P_{\text{Right}}}{P_{\text{Left}}}$

11.(A)

12.(D) The compound (X) is an unsymmetrical anhydride because it produces two different carboxylic acids on hydrolysis. An unsymmetrical anhydride $\text{RCOOCOR}'$ on reduction with excess of LiAlH_4 yields two primary alcohols RCH_2OH and $\text{R}'\text{CH}_2\text{OH}$. The only primary alcohol that gives positive iodoform test is $\text{CH}_3\text{CH}_2\text{OH}$. So, the compound (Y) is $\text{CH}_3\text{CH}_2\text{OH}$. Obviously, the other 1° alcohol (Z) is $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ because the total number of carbon atoms is five. Thus, the structure of the compound (X) is $\text{CH}_3\text{COOCOCH}_2\text{CH}_3$. On hydrolysis, it produces acetic acid and propionic acid.



1.(3) Number of compound which can be reduced by $\text{NaBH}_4 = 3$

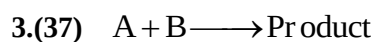


2.(5) $k_1 = \frac{2.303}{t_1} \log \frac{100}{3.125}$ for 96.875% (C) reacted

$k_2 = \frac{2.303}{t_2} \log \frac{100}{50}$ for 50% (A) reacted

Since $t_2 = t_1$

$\therefore \frac{k_1}{k_2} = \frac{\ln 32}{\ln 2} = 5$



$-\frac{d[A]}{dt} = k[A]$

$-\frac{d[A]}{dt} = 1.85 \times 10^{-4} \times [A] \quad \dots (i)$

Assuming the reaction to be of second order

$-\frac{d[A]}{dt} = k'[A][B]$

$-\frac{d[A]}{dt} = k'[A][0.1] \quad \dots (ii)$

Dividing Eq. (i) by (ii), we get

$1 = \frac{1.85 \times 10^{-4}}{k' \times (0.1)}$

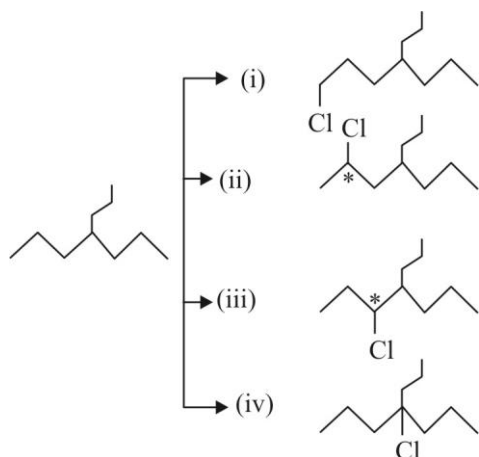
$\therefore k' = 1.85 \times 10^{-3} \text{ L mol}^{-1} \text{ s}^{-1}$

rate = $1.85 \times 10^{-3} \times 0.01 \times 0.02$

$= 3.7 \times 10^{-7}$

$= 37 \times 10^{-8}$

4.(6)



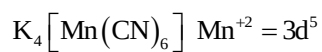
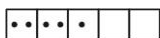
(A) Total structural monochloro product = 4

(B) Total isomer (including stereo) = 6

5.(34) Number of homopolymer = 3 polythene, PVC, Teflon

Number of copolymer polymer = 4

6.(1.73)

CN⁻ = strong fieldUnpaired e⁻ = 1Magnetic moment = $\sqrt{1 \times 3} \text{ BM}$

= 1.732 BM

MATHEMATICS

1.(AB)

$$9\cos^{12}x + \cos^2 2x + 1 - 6\cos^6 x \cos 2x - 6\cos^6 x + 2\cos 2x = 0$$

$$\Rightarrow (3\cos^6 x - 1 - \cos 2x)^2 = 0 \quad \Rightarrow \quad \cos^2 x (3\cos^4 x - 2) = 0$$

$$\Rightarrow \cos x = 0$$

$$x = n\pi + \frac{\pi}{2}, n \in I \quad \text{Or} \quad \cos^4 x = \frac{2}{3}$$

$$\cos x = \pm \sqrt[4]{\frac{2}{3}}; \quad x = m\pi \pm \cos^{-1} \sqrt[4]{\frac{2}{3}}, m \in I$$

2.(ABD)

Let any point on the line through P be $x = \alpha + r \cos \theta, y = 2 + r \sin \theta$.

Put this in $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and $xy = 0$.

$$PA \times PD = PB \times PC \quad \Rightarrow \quad 2\alpha \sin 2\theta + 5 \cos 2\theta = 13$$

$$\Rightarrow 13 \leq \sqrt{4\alpha^2 + 25} \quad \Rightarrow \quad |\alpha| \geq 6$$

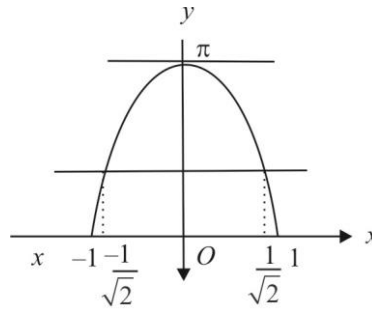
3.(AC)

Differential coefficient of $f(x)$ w.r.t.

$$\sin^{-1} x = -2 \frac{x}{|x|} \text{ at } x = \frac{1}{2} \text{ it is } -2$$

$$f'(0^+) = -2; f'(0^-) = 2$$

$$\Rightarrow f \text{ is not derivable at } x = 0$$



$$4.(ABC) \quad f(x) = \frac{\sin x}{x}$$

5.(ABCD)

Equation of tangent at P is $x + y = 4$

Clearly mirror image of $R(-5, 5)$ lies on line PQ .

Now mirror image R' or R

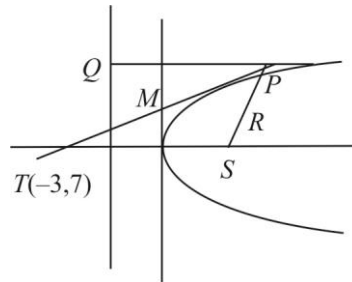
$$\Rightarrow \frac{\alpha' + 5}{1} = \frac{\beta' - 5}{1} = \frac{-2(-5 + 5 - 4)}{2} = 4$$

$$\Rightarrow (\alpha', \beta') = (-1, 9)$$

Let PM cuts the axis at T ; As M is midpoint of PT

$$\Rightarrow T \text{ is } (-3, 7)$$

We know that $SP = ST$ and $\angle SMP = \frac{\pi}{2}$



$$\text{Equation } SP \equiv y - 3 = -\frac{1}{3}(x - 1)$$

$$\Rightarrow x + 3y - 10 = 0$$

$$\text{Let } S \equiv (10 - 3\beta, \beta)$$

$$\text{Again } TS \parallel PQ \Rightarrow \frac{\beta - 7}{13 - 3\beta} = \frac{9 - 3}{-1 - 1} = -3$$

$$\therefore \text{ focus is } (-2, 4)$$

6.(AD)

$$h'(x) = f(x) \Rightarrow h''(x) = f'(x)$$

$$h(1) = 0, f(1) = f'(1) = h'(1) = h''(1) = g(1)$$

$$f(g(x)) = x$$

$$f'(g(x)) \cdot g'(x) = 1 \Rightarrow g'(1) = 1$$

$$G(x) = x^2 \cdot g(x) - x \cdot h(g(x))$$

$$G'(x) = x^2 g'(x) + 2x \cdot g(x) - h(g(x)) - x \cdot h'(g(x)) \cdot g'(x)$$

$$= 2x \cdot g(x) - h(g(x))$$

$$G''(x) = 2 \cdot g(x) + x \cdot g'(x)$$

$$G'(1) = 2g(1) - h(g(1)) = 2g(1) - h(1) = 2$$

$$G''(1) = 2 \cdot g(1) + g'(1) = 3$$

$$7.(C) \quad y = mx \pm \sqrt{a^2 m^2 + b^2} ; \quad k = mh \pm \sqrt{a^2 m^2 + b^2}$$

$$(k - mh)^2 = a^2 m^2 + b^2$$

$$m^2(h^2 - a^2) - 2mhk + k^2 - b^2 = 0 \quad \therefore (m_1 = \tan \theta_1, m_2 = \tan \theta_2)$$

$$m_1 m_2 = \frac{k^2 - b^2}{h^2 - a^2} = \tan \theta_1 \tan \theta_2 = 4 \quad \Rightarrow \quad \frac{y^2 - b^2}{x^2 - a^2} = 4 \Rightarrow \left(\frac{y - b}{x - a} \right) = 4 \left(\frac{x + a}{y + b} \right)$$

$$8.(A) \quad m_1 + m_2 = \frac{2hk}{h^2 - a^2}$$

$$\text{And } \cot \theta_1 + \cot \theta_2 = \lambda$$

$$\Rightarrow \frac{1}{\tan \theta_1} + \frac{1}{\tan \theta_2} = \lambda$$

$$\Rightarrow \frac{\tan \theta_1 + \tan \theta_2}{\tan \theta_1 \tan \theta_2} = \lambda \Rightarrow \frac{\frac{2hk}{h^2 - a^2}}{\frac{k^2 - b^2}{h^2 - a^2}} = \lambda$$

$$\Rightarrow 2hk = \lambda(k^2 - b^2) ; \quad 2xy = \lambda(y^2 - b^2)$$

$$9.(C) \quad g(t) = \lim_{x \rightarrow 0} (1 + a \tan x)^{t/x}$$

$$g(t) = e^{\lim_{x \rightarrow 0} \frac{1}{x} a \tan x} = e^{\lim_{x \rightarrow 0} t a \frac{\tan x}{x}}$$

$$g(t) = e^{ta} = e^{ta}$$

$$g(x) = e^{ax}$$

$$\because a = 2, g(x) = e^{2x}$$

$$g(2) = e^4$$

$$10.(C) \quad f(x) = \begin{cases} x e^{a/x}, & x < 0 \\ x + ax^2 - x^3, & x \geq 0 \end{cases}$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x + ax^2 - x^3 = 0$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} x e^{a/x} = 0 \quad (\text{only when } a \geq 0, \text{ otherwise limit does not exist})$$

$$f(0) = 0$$

$$a \in [0, \infty]$$

$$11.(D), \quad 12.(A)$$

$$f'(x) = \int_0^{\pi} \frac{1}{1+x \cos \theta} d\theta = \int_0^{\pi} \frac{1}{1-x \cos \theta} d\theta \quad \left(\because \int_0^{\pi} g(\theta) d\theta = \int_0^{\pi} g(\pi - \theta) d\theta \right)$$

$$\Rightarrow \quad 2f'(x) = \int_0^{\pi} \frac{2}{1-x^2 \cos^2 \theta} d\theta$$

$$f'(x) = \int_0^x \frac{d\theta}{1-x^2 \cos^2 \theta} = 2 \int_0^{\frac{\pi}{2}} \frac{d\theta}{1-x^2 \cos^2 \theta} = 2 \int_0^{\frac{\pi}{2}} \frac{\sec^2 \theta d\theta}{1+\tan^2 \theta - x^2} = 2 \int_0^{\infty} \frac{dt}{t^2 + 1 - x^2}$$

$$= 2 \frac{1}{\sqrt{1-x^2}} \left[\tan^{-1} \frac{t}{\sqrt{1-x^2}} \right]_0^{\infty} = \frac{\pi}{\sqrt{1-x^2}} \quad \Rightarrow \quad f(x) = \pi \sin^{-1} x + k$$

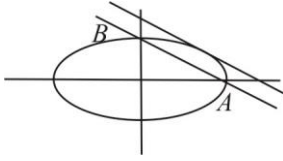
$$\text{But } f(0) = \int_0^{\pi} \frac{\ln 1}{\cos \theta} \therefore f(x) = \pi \sin^{-1} x$$

$$\text{Range of } f(x) = \left(-\frac{\pi^2}{2}, \frac{\pi^2}{2} \right) \left(\because \text{range of } \sin^{-1} x = \left(-\frac{\pi}{2}, \frac{\pi}{2} \right) \right)$$

$f(x)$ is differentiable in the interior of its domain and $f'(x) = 0$ has no solution.

Hence $f(x)$ has no critical points

- 1.(3) Let perpendicular distance of P from the line be h



$$\frac{1}{2} \times h \times 5 = 6(\sqrt{2} - 1) \quad (\text{as } \Delta PAB = 6(\sqrt{2} - 1))$$

$$\Rightarrow h = \frac{12(\sqrt{2} - 1)}{5}$$

Now distance of tangent parallel to AB i.e. $4y + 3x = 12\sqrt{2}$, from line AB is $\frac{12(\sqrt{2} - 1)}{5}$

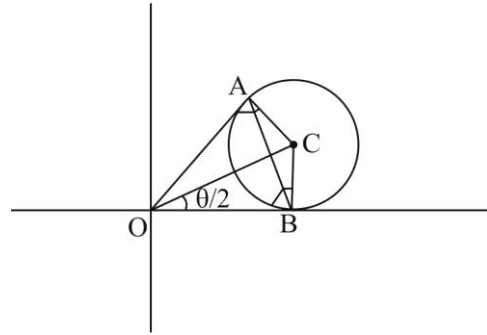
There are just three such points.

- 2.(3) Let C be the centre of the circle & let $\angle AOB = \frac{\theta}{2}$

$$\text{In } \Delta OAC, \cos \frac{\theta}{2} = \frac{2}{\sqrt{5}}$$

$$\text{In } \Delta ABC, a \cos \frac{\theta}{2} = \frac{3}{2}$$

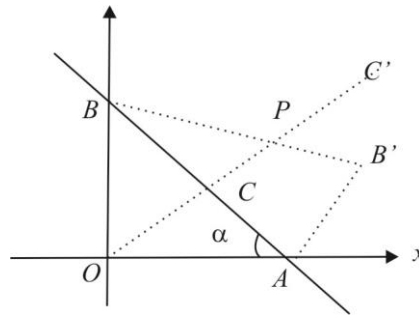
$$\Rightarrow \frac{4a}{\sqrt{5}} = 3$$



- 3.(2) $A \equiv (1, 0), B \equiv (0, \tan \alpha), C \left(\frac{1}{2}, \frac{\tan \alpha}{2} \right)$,

$$B' = \left(\frac{2 \tan^2 \alpha}{1 + \tan^2 \alpha}, \frac{\tan^3 \alpha - \tan \alpha}{1 + \tan^2 \alpha} \right)$$

$$\frac{\Delta ABB'}{\Delta BB'C'} = 2$$



- 4.(2) 2nd equation has solution

$$|x| = \frac{1}{2} \quad \text{or} \quad |y| = \frac{1}{a}$$

Total possible solutions

$$|x| = \frac{1}{2}, |y| = \frac{3}{2} \quad \{4 \text{ points}\}$$

$$|x| = \frac{1}{a} - 1, |y| = \frac{1}{a} \quad \{4 \text{ points}\}$$

$$|x| = \frac{1}{a} + 1, |y| = \frac{1}{a} \quad \{4 \text{ points}\}$$

For 8 possible points

$$\frac{1}{a} = \frac{3}{2} \Rightarrow 3a = 2$$

5.(5)

$$\text{Let } \vec{d} \cdot \vec{a} = \cos y [\vec{a} \vec{b} \vec{c}] = -\vec{d} \cdot (\vec{b} + \vec{c}) \quad [\text{as } \vec{d} \cdot (\vec{a} + \vec{b} + \vec{c}) = 0]$$

$$\Rightarrow \cos y = -\frac{\vec{d} \cdot (\vec{b} + \vec{c})}{[\vec{a} \vec{b} \vec{c}]} \quad \dots(1)$$

$$\text{Similarly } \sin x = \frac{\vec{d} \cdot (\vec{b} + \vec{a})}{[\vec{a} \vec{b} \vec{c}]} \quad \dots(2)$$

$$\Rightarrow 2 = -\frac{\vec{d} \cdot (\vec{a} + \vec{c})}{\vec{a} \vec{b} \vec{c}} \quad \dots(3)$$

Adding these we get $\sin x + \cos y + 2 = 0$

$$\Rightarrow \sin x + \cos y = -2$$

$$\Rightarrow \sin x = -1, \cos y = -1$$

$$\Rightarrow x = (4n-1)\frac{\pi}{2}, y = (2n+1)\pi$$

Since we want minimum value of $x^2 + y^2$, so $x = -\frac{\pi}{2}, y = \pi$

$$\Rightarrow x^2 + y^2 = \frac{5\pi^2}{4} \Rightarrow \lambda = 5$$

6.(5)

$$\text{Equation of normal at point } T\left(t, \frac{t^2}{2}\right) \text{ to } C \text{ is } y - \frac{t^2}{2} = \frac{-1}{t}(x - t) \Rightarrow y = \frac{-x}{t} + \frac{t^2}{2} + 1$$

If circle S touches curve C at T .

$$\Rightarrow (x(t), y(t)) \text{ must lie on normal}$$

$$\Rightarrow y(t) = \frac{-x(t)}{t} + \frac{t^2}{2} + 1$$

Also distance of centre of circle from T equals radius = distance of centre of circle from x-axis.

$$\Rightarrow (x(t) - t)^2 + \left(y(t) - \frac{t^2}{2}\right)^2 = (y(t))^2$$

$$x(t) = \frac{t(1 + \sqrt{t^2 + 1})}{2}, y(t) = \frac{t^2 + 1 - \sqrt{t^2 + 1}}{2}$$

$$\Rightarrow x(t)y(t) = \frac{1}{4}t^3\sqrt{t^2 + 1}; \quad \lim_{r \rightarrow 0} \int_r^1 \frac{1}{4}t^3\sqrt{t^2 + 1}dt$$

$$\text{Let } \sqrt{1 + t^2} = z \Rightarrow 2tdt = 2zdz \quad \Rightarrow \quad \lim_{t \rightarrow 0^+} \int_{\sqrt{1+t^2}}^{\sqrt{2}} \frac{1}{4}(z^2 - 1)z^2dz = \frac{\sqrt{2} + 1}{30}$$